Instructions for Academic Project Report (Computer Networks)

Follow the following instructions to draft the project report:

- **General Requirements**
  - **Report length**: 15–20 pages (excluding references/appendices)
  - **Font**: Times New Roman, size 12 pt
  - **Line spacing**: 1.5
  - **Margins**: 1 inch (all sides)
  - **Content Alignment**: Justified
  - **Page numbering**: Bottom center or right
  - Use formal academic language (no slang or informal tone)

- **Structure of the Report (Mandatory Sections)**
  - **Title Page**
    - Project title
    - Student name(s) and roll number(s)
    - Course name (Computer Networks)
    - Instructor name
    - Institution name
  - **Introduction**
    - Background of the problem
    - Motivation
    - Scope of the project
    - Applications
  - **Problem Statement & Objectives**
    - Clearly define:
      - What problem you are solving
      - Why it is important
    - List specific objectives
  - **System Design / Methodology**

    Depending on project type:

    For protocol-based:

    - ✓ Protocol design
    - ✓ Packet structure
    - ✓ State diagrams

For client-server applications:

- ✓ Architecture diagram
- ✓ Communication flow
- ✓ APIs / sockets used

For network traffic applications:

- ✓ Data flow
- ✓ Traffic analysis method

Include:

- ✓ Diagrams (architecture, flowcharts)
- ✓ Algorithms (if applicable)

- **Implementation Details**
  - Tools/technologies used (e.g., Python, Java, sockets, Wireshark, GNS3, etc.)
  - Key modules
  - High Quality Screenshots (if applicable)
- **Results and Analysis**
  - Output of your system
  - Screen shots
  - Performance metrics if any
- **Discussion**
  - Interpretation of results
  - Strengths and limitations
- **References (Mandatory)**
  - Use IEEE citation style
  - Cite RFCs
  - Minimum 5–10 references
- **Appendix**
  - Code snippets
  - Additional figures if any

- **Figures, Tables, and Equations**
  - All figures must have:
    - Figure number + caption
    - Refer to them in text: "As shown in Figure 2…"
  - Tables should be properly labelled, Refer to them in text.
  - Equations should be numbered (if used)